

Manasa Vijayaraghavan

Data Scientist & ML Engineer

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SUMMARY

Data Scientist and ML Engineer pursuing an M.S. in Applied Data Science at USC, with hands-on experience building end-to-end ML pipelines, GenAI workflows, and production-grade analytics systems. Filed a patent for real-time sign-language translation AI and delivered measurable impact at Ford Motor Company and iNextLabs. Open to Data Science, Data Analytics, AI Engineering, ML Engineering, and Data Engineering roles.

EXPERIENCE

Wipro — AI Native Intern

June 2026 – August 2026

- Built **AgenticOS BFSI Skills**, an agentic decision-support library for **Banking, Financial Services & Insurance**, owning two life-insurance skills end to end.
- Architected multi-step **LangGraph** agent workflows with a **human-in-the-loop** gate, enforcing **maker-checker approval** on every run.
- Engineered deterministic **Python scoring engines** for underwriting tiers and fraud detection, confining the LLM to extraction and narration to eliminate hallucination on key figures.
- Implemented real document ingestion (**PDF/XLSX**) and wired skills to data stores over read-only **MCP servers**.
- Validated with **golden-case regression suites** and an examiner-traceable audit trail on every run.

Ford Motor Company — Data Science Intern

Feb 2025 – May 2025

- Built a **multimodal emotion analysis** system using **Python, DeepFace** (facial recognition via OpenCV), **VADER** (text sentiment), and **Vosk** (speech-to-text); fused video, audio, and text modalities into a unified emotion classification, deployed on **GCP** with **Vertex AI**.
- Collaborated with a cross-functional corporate AI team applying **NLP, computer vision**, and **speech processing** to real-world multimodal data, improving system efficiency by **75%**.

iNextLabs — Data Science Intern

Aug 2024 – Oct 2024

- Built **NLP, anomaly detection, and GenAI** solutions using **PyTorch** and **Azure AI**; integrated into production systems, reducing issue resolution time by **40%**.
- Designed and deployed **AI analytics pipelines** and dashboards serving **500+ users** across technical and non-technical teams.

PROJECTS

SignMate AI. Patented real-time CNN system translating Indian Sign Language to text (PyTorch, OpenCV). Patent filed with IP India; published on the IP India website.

Finance Analytics & NL→SQL Intelligence. ETL pipelines for financial data with forecasting models and a PostgreSQL + pgvector backend; LangChain-powered plain-English SQL lets non-technical users query data directly.

Production MLOps Platform & LLM Inference. Full ML lifecycle — MLflow experiment tracking, DistilBERT inference, drift detection, and zero-downtime rollouts on Docker + GCP Cloud Run.

Large-Scale Customer Churn Prediction. PySpark pipeline on 1M+ rows — 92.6% AUC, 45 engineered features, K-Means segmentation, and SHAP explainability with a Plotly dashboard.

Local RAG Pipeline — LangChain, Ollama & FAISS. Privacy-first retrieval-augmented generation, 100% local with zero API cost; sub-second retrieval over 10K+ document chunks.

Fraud Detection & Churn Prediction — ML Research. Automotive ECU fraud detection at 98.7% AUC on a 2% imbalanced rate; PyTorch autoencoder embeddings for churn across 4 benchmarked architectures.

SKILLS

Languages	Python, SQL, Git / GitHub
ML / AI	Machine Learning, NLP, LLMs & RAG, Computer Vision, Prompt Engineering
Frameworks	PyTorch, TensorFlow, Scikit-learn, LangChain, HuggingFace, XGBoost
Data Engineering	ETL Pipelines, PySpark / Spark, Feature Engineering, Data Cleaning & EDA
Visualization	Tableau, Power BI, Matplotlib / Seaborn
Cloud	GCP, Azure ML, AWS, Docker, PostgreSQL

EDUCATION

University of Southern California — M.S. Applied Data Science	May 2027 · Los Angeles, CA
SRM Institute of Science and Technology — B.Tech. Computer Science Engineering	Jun 2025 · Chennai, India

SELECTED PUBLICATIONS & CERTIFICATIONS

- Patent Filed — SignMate AI (IP India) · Real-time sign-language translation via CNNs
- Publication — Project Eye (TechRxiv, peer-reviewed) · Neural ML vision systems
- AIVerify: An Enhancement of AI Security (Medium) · IMDA Singapore Project Moonshot
- Predictive Model for Cancer Detection (Medium) · Healthcare ML classification
- Google Cloud AI (Advanced) · Azure ML Engineer (Microsoft)